6 TESTS OF HYPOTHESES

INTRODUCTION

A hypothesis is an assumption about the population parameter to be tested based on sample information.

Procedure of Hypothesis Testing

The general procedure followed in testing hypothesis comprises the following steps:

(1) Set up a hypothesis

The first step in hypothesis testing is to establish the hypothesis to be tested. Since statistical hypothesis are usually assumptions about the value of some unknown parameter, the hypothesis specifies a numerical value or range of values for the parameter. The conventional approach to hypothesis testing is not to construct single hypothesis about the population parameter, but rather to set up two different hypothesis. These hypothesis are normally referred to as

(i) null hypothesis denoted by H_0 - A hypothesis which states that there is no difference between the assumed and actual value of the parameter is the null hypothesis. A hypothesis that is different from the null hypothesis is the alternative hypothesis.

(ii) Alternative hypothesis denoted by H_1 . the alternative hypothesis specifies those values that the researcher believes to hold true.

If the sample information leads us to reject H_0 , then we will accept the alternative hypothesis H_1.

(2) Set up a suitable significance level

Select a suitable level of significance usually denoted by "α". The confidence with which an experimenter rejects or retains null hypothesis depends on the significance level adopted. When the null hypothesis is rejected at $\propto = 0.05$, the test result is said to be "significant". When the null hypothesis is rejected at $\propto = 0.01$, the test result is said to be "highly significant".

(3) Determination of a suitable test statistic

The third step is to determine a suitable test statistic and its distribution. Many of the test statistics that that we shall encounter will be of the following form:

$$\text{Test statistic} = \frac{Sample\ \ Statistic\ \ - Hypot\ hesised\ \ population\ \ \ parameter}{standard\ \ \ error\ \ of\ \ the\ sample\ \ ststistic}$$

(4) Determine the critical region

It is important to specify, before the sample is taken, which value of the test statistic will lead to a rejection of H_0.The former is called the critical region. The value of α , the level of significance, indicates the importance that one attaches to the consequences associated with incorrectly rejecting H_0 . It can be shown that the level of significance is α , the optimal critical region for a two sided test consists of that $\propto/2$ per cent in the right- hand tail of the distribution plus that $\propto/2$ per cent in the left hand tail. Thus, establishing a critical region is similar to determining a 100 (1- $\propto$)% confidence interval. In general,

one uses a level of significance of $\propto = 0.05$, indicating that one is willing to accept a 5 per cent chance of being wrong to reject H_0.

(5) Doing Computations

The fifth step in testing hypothesis is the performance of various computations from a random sample of size n, necessary for the test statistic obtained in step (3). Then, we need to see whether sample result falls in the critical region or in the acceptance regions.

(6) Making decisions

Finally, we may draw statistical conclusions and the management may take decisions. A statistical decision or conclusion comprises either accepting the null hypothesis or rejecting it. The decision will depend on whether the computed value of the test criterion falls in the region of rejection or the region of acceptance. If the hypothesis is being tested at 5 per cent level of significance and the observed set of results has a probability less than 5 per cent, we reject the null hypothesis and the difference between the sample statistic and the hypothetical population parameter is considered to be significant. On the other hand, if the testing statistic fall in the region of non-rejection, the null hypothesis is accepted and the difference between the sample statistic and the hypothetical population parameter is not regarded as significant, i.e. it can be explained by chance variations.

Type I and Type II Errors

When a statistical hypothesis is tested, there are four possible results:

1) The hypothesis is true but our test rejects it.
2) The hypothesis is false but our test accepts it.
3) The hypothesis is true but our test accepts it.
4) The hypothesis is false and our test rejects it.

Obviously, the first two possibilities lead to errors. If we reject a hypothesis when it should be accepted (possibility No. 1), we say that a *Type 1 error* has been made. On the other hand, if we accept a hypothesis when it should be rejected (possibility No.2), we say that a *Type 11error* has been made. In either case a wrong decision or error in judgement has occurred.

TWO KINDS OF ERRORS IN HYPOTHESIS TESTING

Decision	Condition: H_0: True	Condition: H_o: False
Accept H_0	Correct decision	Type 11 Error
Reject H_0	Type 1 Error	Correct Decision

The probability of committing a type 1 error is designated as " $\propto$" and is called the level of significance.

Therefore,

$$\propto = P_r \text{ [Type 1 error]}$$

$$= p_r \text{ [Rejecting } H_0/H_0 \text{ is true]}$$

Must be the complement of

$$(1-\alpha) = p_r \text{ [Accepting } H_0/H_0 \text{ is true].}$$

This probability $(1-\alpha)$ corresponds to the concept of $100(1-\alpha)\%$ confidence interval. Our efforts would obviously be to have a small probability of making a type 1 error. Hence the objective is to construct the test to minimize $\propto$.

Similarly, the probability of committing a type 11 error is designated by β. Thus

$\beta = P_r$ [Type 11 error]

$= p_r$ [Accepting H_0 / H_0 is false]

and $(1-\beta) = p_r$ [Rejecting H_0 / H_0 is false]

This probability $(1-\beta)$ is known as the power of a statistical test.

The following table gives the probabilities associated with each of the four cells shown in the previous table:

The decision is:	The null hypothesis is	
	True	False
Accept H_0	$(1-\alpha)$ Confidence level	β
Reject H_0	α	$(1-\beta)$ Power of the test
Sum	1.00	1.00

Note that the probability of each decision outcome is a conditional probability and the elements in the same column sum to 1.0, since the events with which they are associated are complement. However, α and β are not independent of each other, nor are they independent of the sample size n. When n is fixed, if α is lowered then β normally rises and vice versa. If n is increased, it is possible for both α and β to decrease. Since increasing the sample size involves money and time, therefore, one should decide how much additional money and time , he is willing to spare on increasing the sample size in order to reduce the size of α and β

.

In order for any tests of hypothesis or rules of decisions to be good, they must be designed so as to minimize errors of decision. However, this is not as simple matter, since for a given sample size, an attempt to decrease one type of error is accompanied in general by an increase in other type of error. The probability of making type 1 error is fixed in advance by the choice of level of significance employed in the test. We can make the type 1 error as small as we please, by lowering the level of significance. But, by doing so, we increase the chance of accepting a false hypothesis, *i.e.,* of making a type II error. It follows that it is impossible to minimize both errors simultaneously. In the long run, errors of type 1 are perhaps more likely to prove serious in research programmes in social sciences than are errors of type II. In practice, one type of error may be more serious than the other and so a compromise should be reached in favour of limitations of the more serious error. The only way to reduce both types of error is to increase the sample size which may not be possible.

One –Tailed and Two Tailed Tests

Basically, there are three kinds of problems of tests of hypothesis. They include:

i. Two -tailed tests ii. Right- tailed test and (iii) Left- tailed test.

Two- tailed test is that where the hypothesis about the population mean is rejected for value of falling into either tail of the sampling distribution. When the hypothesis about population mean is rejected only for value of falling into one of the tails of the sampling distribution, then it is known as one-tailed test. If, it is right tail then it is called right -tailed test or one–sided alternative to the right and if it is on the left tail, then it is one-

sided alternative to the left called left-tailed test. For example, $H_0 : \mu = 100$ tested against $H_1 : \mu > 100 \; or \; < 100$ is one tailed test since H_1 specifies that μ lies on particular side of 100. The same null hypothesis tested against $H_1 : \mu \neq 100$ is a two tailed test since μ lies on a particular side of 100. The following diagram would make it clearer:

The following table gives critical values of z for both one-tailed and two tailed tests at various levels of significance. Critical values of z for other levels of significance are found by use of the table of normal curve areas.

Level of significance	0.10	0.05	0.01	0.005	0.0002
Critical value of z for one tailed tests	-1.28 or 1.28	-1.645 or 1.645	-2.33 or 2.33	-2.58 or 2.58	-2.88 or 2.88
Critical value of z for two tailed tests	-1.645 and 1.645	-1.96 and 1.96	-2.58 and 2.58	-2.81 and 2.81	-3.08 and 3.08

Tests of Hypothesis Concerning Large Samples

Though it is difficult to draw a clear cut line of demarcation between large and small samples, it is generally agreed that if the size of sample exceeds 30, it should be regarded as large sample. The tests of significance used for large samples are based on the following assumptions:

1) The sampling distribution of a sample statistic is approximately normal.

2) Values given by the samples are sufficiently close to the population value and can be used in its place for the standard error of the estimate.

Thus, we have seen that the normal distribution plays a vital role in tests of hypothesis based on large samples (central limit theorem).

Suppose $\hat{\theta}$ is an unbiased estimate of θ, the population parameter. On the basis of $\hat{\theta}$, taken from sample observations, we want to test the hypothesis whether the sample is drawn from a population whose parameter value is θ, i.e , we have to test the hypothesis:

$H_0 : \theta = \hat{\theta}$

If sampling distribution of θ is normal then,

$$z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \sim N(0,1)$$

Standard Normal Distribution compared with distribution when n = 5 and $n = 15$

The t-distribution has different shapes depending on the size of the sample. When the sample is quite small, for example, *if n* equals five, the height of the t-distribution is shorter than the normal distribution and the tails are wider. As *n* nears 30, however, the t-distribution approaches the normal distribution in shape.